

SACHIN K SUNNY

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SUMMARY

Results-driven Software Engineering Manager with a strong **expertise in microservice architecture, backend systems, application security and data engineering**. Proven track record of leading engineering teams and architecting scalable, high-impact systems on GCP. Key accomplishments include architecting the Canada e-commerce platform modernization (\$4M annual revenue), spearheading a cloud cost optimization initiative **saving over \$2.4M annually**, and delivering a portfolio of projects generating millions in revenue (Item Substitutions **\$2.5M**, Birthday Party Booking **\$2.1M**).

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Information Systems Management: Business Intelligence & Data Analytics

August 2020

Courses: Machine Learning, NoSQL Databases, Data Structures, Unstructured Data Analytics

GPA: 4.10/4.33

Indian Institute of Technology, Madras

July 2016

Dual Degree: Bachelor of Technology & Master of Technology in Engineering Design

GPA: 3.88/4.00

SKILLS, TECHNOLOGIES & PATENTS

Languages & Tools: Java, Python, R, Excel, Tableau, HTML, CSS, JavaScript, Node.js, React, REST APIs, Spring Boot

Database & Big Data: SQL, PostgreSQL, Google Cloud Spanner, Redis, MongoDB, Cassandra, Neo4j, Elasticsearch

Cloud Technologies & DevOps: Google Kubernetes Engine (GKE), Google Cloud Platform (GCP), Google BigQuery, Dataflow, GCP Cloud Run, Jenkins, Bitbucket, GitHub, Akamai, Kong API Gateway, Prometheus, Grafana, Datadog

Functional Skills: Predictive Analytics, Database Management, A/B Testing, Data Visualization, Microservice Architecture

Relevant Technical Training & Coursework: Advanced System Design, Generative AI for Business Leaders, Application Security, Natural Language Processing (BERT, IBM Watson), Big Data, Data Analytics, AWS Fundamentals

Patents: Piston mechanism in Vacuum Drying Washing Machine (5973/CHE/2014), Nov 2014

PROFESSIONAL EXPERIENCE

Michaels, Manager: Software Engineering

Irving, TX: May 2023 - Present

- **Project Delivery**: Spearheaded a team of 5 to deliver **Item Substitutions (\$2.5M top line)**, **Birthday Party Booking (\$2.1M annualized GMV)**, **Affirm Payment Option (\$1.2M top line)**, SMS Vendor Replacement (\$555k savings), Same Day Delivery 2.0 (\$536k shipping cost reduction), and Extend Integration (\$400k in revenue share) on schedule
- **Canada Re-Platform**: Architected modernization of the Canada e-commerce platform, driving a seamless migration that contributes **\$4M in annual revenue**. Directed end-to-end migration of critical data (users, orders, returns, cancellations, appeasements, product catalogs), ensuring seamless integration with payment, inventory, fulfillment, reporting, and supply chain systems. Designed a scalable, multi-locale architecture leveraging core US platform components to improve operational efficiency, consistency, and customer experience.
- **Omnichannel Account**: Accelerated omnichannel loyalty by unifying online, in-store, and rewards account ecosystems, **increasing customer account creation rates from ~50% to 85%**. Spearheaded design and implementation of an automated, data-driven enrollment platform using Google Pub/Sub, Google Spanner, and API integrations across retail and digital channels. Streamlined onboarding, strengthened cross-channel data consistency, and drove deeper engagement through unified customer experiences.
- **GCP Cost Optimization**: Championed a cloud cost optimization initiative, **reducing annual infrastructure expenses by over \$2.4M**. Led cross-functional efforts to decommission underutilized resources, implement governance controls, and adopt cost-efficient cloud architectures, enhancing financial discipline and operational efficiency.
- **Customer Care AI**: Directed an end-to-end AI proof of concept, evaluating two conversational AI vendors for voice and chat channels to improve containment, customer satisfaction, and deployment efficiency. Key use cases included *Where Is My Order*, *Returns*, *Cancellations*, and *Internal Help Desk Support*. Achieved **~40% reduction in live agents** and **~65% reduction in customer care operating costs** while accelerating the company's AI-driven service transformation.
- **Log Management Optimization**: Engineered an internal log exploration platform by routing GCP server logs through LogSink into a Filebeat-Logstash-Elasticsearch-Kibana pipeline. Streamlined observability, **reducing monthly log costs by \$20K** and improving scalability and retrieval efficiency. Audited log severity levels, removing redundant WARNING and ERROR entries to reduce noise and storage overhead.

- **Application Performance Monitoring:** Spearheaded adoption of Datadog to enhance service instrumentation and end-to-end observability, giving developers and SRE teams deeper visibility and faster issue resolution. Built tailored Grafana dashboards powered by Prometheus to track system load, latency, and error rates, while implementing real-time alerts for business-critical flows, elevating operational awareness and containing incident impact.
- **Application Security Enhancements:** Strengthened platform security by **patching 100+ vulnerabilities** and **refactoring 15+ microservices in 1.5 months** to align with secure coding standards and compliance requirements. Upgraded dependencies to secure versions, implemented input sanitization and parameterized queries to prevent SQL injection and XSS attacks, enforced API rate limits and gateway throttling to mitigate DDoS risks, and applied least-privilege access controls. These efforts reduced the attack surface, improved service reliability, and reinforced overall application resilience.

Michaels, Software Development Engineer Backend III

Irving, TX: October 2020 - May 2023

- **Data Engineering & System Design:** Designed a **multi-threaded, in-memory cached, scalable, fault-tolerant**, and automated data pipeline to **migrate ~500,000 item details** (300+ fields in English & Canadian French) to GCP hosted MongoDB collection, improving data reliability and availability.
- **Platform Migration:** Collaborated with cross-functional teams to translate requirements into code, enabling successful launch of the self-checkout app and re-platforming of legacy US & B2B sites within a year **adding ~\$1B to the top line**.
- **REST API Development:** Built Spring Boot JPA REST endpoints to provide six teams fast access to item attributes and pricing in indexed MongoDB collections, **reducing average API response time from 150ms to 30ms**.

Credit Suisse, Market Risk Analyst

Mumbai, India: July 2016 - July 2019

- **Modelling:** Built Risk Not in VaR (RNIV) models in MS Excel to alter the methodology according to the latest regulatory rules and ascertain the non-modellable risk factors responsible for the market risk **capital of \$1 Billion**.
- **Database & strategy:** Retrieved data from the market risk database, designed a tail risk contribution model to **allocate \$750M in capital** and identified hedging strategies to achieve a **reduction of \$150M in capital**.
- **Scripting & automation:** Combined XML Scripts and VBA Code in MS Excel which **improved accuracy from ~80% to ~98%** and **cut down analysis time from 15 to 3 days (80% reduction)** for high risk factor identification to advise risk-reduction strategy.

ACADEMIC PROJECTS

Emerging Artist identification from social media engagement in YouTube (Client: PEX)- Capstone May - Aug 2020

- **Clustering:** Separated historical data into **4 distinct growth rate clusters** using KernelKMeans clustering at video level
- **Feature Engineering:** Performed data cleaning and imputation to calculate days to reach view milestones.
- **Modelling:** Applied SMOTE for imbalanced classification and trained an XGBoost model, achieving ROC AUC of **0.783**, enabling identification of emerging artists for targeted recommendations.

Machine Learning

Jan - May 2020

- **Reinforcement Learning:** Implemented Q-Learning with linear function approximation and ϵ -greedy exploration (using NumPy alone) on a coarse-coded state space (2048 indices) to train an agent to move a car uphill.
- **Hidden Markov Models:** Applied forward-backward algorithm with log-space arithmetic and Minimum Bayes Risk prediction for Named Entity Recognition, achieving **83.3% accuracy**.
- **Neural Networks:** Built a 1-hidden-layer neural network from scratch with tunable number of hidden units, learning rate, and epochs to classify handwritten letters; vectorized code for speed, achieving **85.7% accuracy**.

Distributed Systems for Information Systems Management

Jan - May 2020

- **Android, cloud deployment & database:** Engineered a BookFinder Android app to extract data from GoodReads API, deployed the web service on Heroku, and visualized logs stored in MongoDB via a dashboard.
- **Blockchain & sockets:** Implemented a blockchain simulation with transaction addition, verification, corruption, and repair; modeled as client-server system using JSON messages over TCP sockets.

Unstructured Data Analytics

Jan - May 2020

- **Clustering:** Used TF-IDF vectorizer to create a document matrix and applied Gaussian Mixture Models to classify spam vs. ham emails.
- **Model Tuning:** Optimized kNN, SVM, and Random Forest classifiers through hyperparameter selection, achieving an F1 score of 0.94.